Journal of Network and Computer Applications 35 (2012) 1162–1174

Contents lists available at SciVerse ScienceDirect

Journal of Network and Computer Applications

journal homepage: www.elsevier.com/locate/jnca

Performance improvements for network-wide broadcast with instantaneous network information

Abhik Banerjee*, Chuan Heng Foh, Chai Kiat Yeo, Bu Sung Lee

Centre for Multimedia and Network Technology, School of Computer Engineering, Nanyang Technological University, Singapore

ARTICLE INFO

Article history:
Received 27 April 2011
Received in revised form
21 November 2011
Accepted 19 January 2012
Available online 28 January 2012

Keywords:
Network-wide broadcasting
Wireless ad hoc networks
Stateless broadcasting algorithms
Network information

ABSTRACT

We study the performance of network-wide broadcasting as a function of the information implicitly available at nodes from neighbourhood transmissions. We term this set of instantaneous information as *network information*. Our discussion is focused on stateless broadcasting algorithms in which nodes decide on their forwarding behaviour based on the available *network information*. While stateless broadcasting schemes in the existing literature use various design guidelines that take advantage of specific aspects of the information, we develop a unified analytical model by characterizing the information available during different stages of broadcasting. Thus, our results are applicable to all stateless algorithms. We analyze broadcasting performance in terms of the transmission probability and redundancy of transmissions. Subsequently, we use our results to obtain insights on the feasibility conditions governing algorithm design depending on the network density and costs. While the first part of the work considers ideal channel conditions modeled as a unit disk graph (UDG), we subsequently enhance the model using a quasi-unit disk graph model (QUDG) to understand the effect of dynamic channel conditions.

1. Introduction

Network-wide broadcast is a crucial operation which is fundamental to a wide range of wireless network applications. Essential network functions such as routing make use of network-wide broadcasting for purposes of route establishment for reactive routing in ad hoc networks (Perkins et al., 2003). It is also essential for other applications such as information dissemination in sensor networks and for localization related tasks.

Therefore, optimized design of broadcasting algorithms is crucial to overall network performance. A straightforward way to broadcast a packet is by blind flooding. However, a multitude of issues with blind flooding in dense networks has been identified as the broadcast storm problem by Ni et al. (1999). Increase in network size and density results in a higher number of transmissions due to flooding, which leads to redundancy and incurs contention and collision overheads. This necessitates the need for design of broadcast algorithms that optimize the set of transmissions.

In this paper, we are motivated to study the performance benefits achievable for network-wide broadcasting without incurring protocol overheads. As broadcasting often forms a part of other network applications, it is crucial that minimum overheads are incurred to ensure performance guarantees for the final application.

1.1. Motivation

A majority of research in broadcast algorithm design focuses on identifying a minimum connected dominating set (MCDS) across the network. As determination of an optimal MCDS requires knowledge of the global network topology, distributed algorithms seek to optimize broadcasting of packets based on local neighbourhood information. However, dissemination of neighbourhood information involves additional communication overheads such as periodic broadcast of hello packets, affecting the overall network performance. The impact on network resources aggravates with increase in node density. Further, it is difficult for such a design to adapt to dynamic topologies. Therefore, it is necessary to consider alternatives that offer robustness of design while impacting minimal network resources.

An alternate class of broadcast algorithms relies on information derived from broadcast packets to determine node forwarding behaviour. They have been referred to as stateless broadcasting algorithms in the existing literature (Heissenbuttel et al., 2006) since

Abbreviation: AR, alerted region; BOP, broadcast optimization phase; BRP, broadcast recognition phase; CDS, connected dominating sets; FR, forwarding region; QUDG, quasi-unit disk graph; RREQ, route request; SRB, savings rebroadcast ratio; MSG, transmitted message; TR, transmission coverage region; UDG, unit disk graph

* Corresponding author. Tel.: +65 822 66 360.
E-mail addresses: abhi0018@ntu.edu.sg (A. Banerjee), aschfoh@ntu.edu.sg (C. Heng Foh), asckyeo@ntu.edu.sg (C. Kiat Yeo), ebslee@ntu.edu.sg (B. Sung Lee).

1084-8045/$ - see front matter
doi:10.1016/j.jnca.2012.01.008

they do not depend on any prerequisite neighbourhood information at nodes. This makes them an ideal choice for purely distributed implementation. The primary design principle of stateless broadcasting algorithms is to maximize the additional coverage of each transmission within its neighbourhood. As it is difficult to accurately estimate the extent of coverage in a neighbourhood, nodes rely on information derived from overheard broadcasts to estimate the expected additional coverage (EAC). The wireless broadcast advantage (WBA) implies that transmission from any node is received by all others within its transmission range. The information implicitly available from a transmission can therefore be utilized by all nodes receiving it. We term this set of implicitly available information as the *network information*.

The primary guidelines for stateless broadcasting algorithms are derived from the observations by Ni et al. (1999) where they analyzed the problems of indiscriminate broadcasting, terming it as the *Broadcast Storm* problem. Stateless algorithms that use *network information* can broadly be classified as distance based and counter based schemes. Distance based schemes are based on the observation that transmissions from nodes located further away from a broadcasting node are likely to maximize the EAC. Counter based schemes build on the observation that a higher number of transmissions in the neighbourhood of a node drastically reduce the EAC. Protocol design based on stateless algorithms attempt to identify optimum distance and counter thresholds that minimize the transmission probability while achieving maximum coverage. To allow protocol designs to adapt to varying topologies, nodes use random or deterministic delays to observe neighbourhood transmission behaviour (Heissenbuttel et al., 2006).

The focus of this paper is to identify the performance benefits achievable for network-wide broadcasting by utilizing only implicitly available *network information*. We obtain a unified performance analysis of stateless broadcasting algorithms by characterizing the *network information* available at a node during different stages of broadcasting. We are motivated by the fact that the resulting performance involves minimum transmission overheads. Further, the instantaneous nature of *network information* implies that algorithm design can adapt to dynamic network topologies. Our characterization of the *network information* takes into consideration salient features of distance and counter based broadcasting schemes, thereby allowing us to concentrate on a generalized view of stateless broadcasting rather than individual algorithms. To our knowledge, ours is the first work that provides such a generalized approach of stateless broadcasting performance analysis.

1.2. Related work

Existing research has characterized the overheads incurred by ad hoc network protocols. Manfredi et al. (2011) analyze the usefulness of maintaining state information depending on network characteristics. The authors in Viennot et al. (2004), Zhou et al. (2003), and Wang and Abouzeid (2008) analyze the overheads involved in reactive and proactive routing protocols for ad hoc networks. The impact of route discovery overheads on the final routing performance has been highlighted by the authors in Viennot et al. (2004) and Zhou et al. (2003), thereby underscoring our motivation for exploring the performance benefits achievable for network-wide broadcasting that incur minimum overheads.

Stateless algorithms are a result of the observations by Ni et al. (1999) that relate effectiveness of broadcasting to the redundancy arising from neighbouring transmissions. Subsequent efforts (Williams and Camp, 2002) have looked to adapt the random delay values used in distance and counter based schemes. Chen et al. (2005) differentiate the delays used by nodes based on their distance from the source. Heissenbuttel et al. (2006) and Heissenbuttel (2005) propose adapting the delay to network conditions. In addition to the observations in Ni et al. (1999), important insights are also gained from the references cited above. Firstly, the authors in Chen et al. (2005) make use of the fact that the additional coverage is a function of both the distance to overheard transmissions as well the number of them. Secondly, since stateless algorithms are distributed, it is necessary for them to adapt to network conditions such as congestion arising from higher contention.

In spite of a rich literature of broadcasting algorithms, analytical treatment of the broadcast performance has been limited. Most of the research on stateless algorithms have focused on probabilistic flooding. Coverage and reliability analysis of blind flooding have been compared to that of probabilistic flooding by Viswanath and Obraczka (2006). Shah-Mansouri et al. (2008) also focus on the analysis of probabilistic flooding, though they emphasize on a non-repetitive flooding scheme. Optimal values of gossiping probability were studied in detail by Haas et al. (2006). More recently, random graph theory results are used to obtain optimal values of forwarding probability in Oikonomou et al. (2010). Jun et al. (2010) obtain the delay performance of flooding using queueing models. Kuo and Liao (2007) analyze the effect of flooding on the hop count. Though their focus is not specifically on flooding, Foh et al. (2007) obtain a distribution of hop lengths in a reactive routing protocol. The closest to our current focus in terms of analyzing the effect of *network information* are Williams et al. (2004) and Zhang and Jiang (2004) which look at distance and counter based schemes. However, the focus of both the analyses is to compare individual schemes and not to evaluate the impact of available information in a generic sense.

1.3. Contributions

As part of our analysis, we first study the performance under ideal channel conditions using a unit disk graph (UDG) model of a network. We identify distinct stages in the broadcasting behaviour of individual nodes and analytically study how forwarding probability of nodes is impacted by the *network information* at each stage. Our analysis methodology looks at how broadcasting performance is affected as the message propagates through the network. We also analyze the redundancy of transmissions and how they compare with the conclusions drawn by Ni et al. (1999). We validate our analytical results with that of simulation results. Our simulation setup involves broadcasting as part of route discovery, which allows us to evaluate other aspects of protocol performance. The key difference of our work with existing analytical studies is that, instead of analyzing individual schemes, we clearly show the impact of *network information* on the performance.

Subsequently, we use our analytical results to draw insights on performance of broadcasting algorithms in general. We discuss how the reliability of broadcasting is affected by the node density in the network. We also obtain feasibility regions within which a broadcast algorithm needs to operate given the network costs it incurs. Lastly, we adapt our model to a realistic approximation of network conditions by using a Quasi-Unit Disk Graph (QUDG) model to replace the UDG model. This allows us to analyze broadcasting reliability in dynamic channel conditions and subsequently draw insights for robust algorithm design.

In summary, the primary contributions of this paper are as follows:

1. A unified performance analysis of stateless broadcasting obtained in terms of the *network information* at different stages of broadcasting.
2. Insights on broadcast algorithm design based on the above model in terms of feasibility conditions for network density and transmission costs.

3. An extension of the above model to account for dynamic channel conditions which therefore provide further insights for robust algorithm design.

1.4. Organization

Section 2 provides the basis for the discussion in the rest of the paper. We first outline the primary objectives of broadcast algorithm design and subsequently identify different stages of broadcasting and define *network information* during each stage. Performance analysis under ideal channel conditions is done in Section 3 and validated using simulation results. In Section 4, the results of the proposed analytical model are used to obtain insights for algorithm design. The performance analysis is extended to account for dynamic channel conditions in Section 5. Section 6 concludes the paper.

2. Background

This section forms the basis of the analysis in the rest of the paper. First, we outline the primary features of broadcast algorithm design that we study. Subsequently, we identify individual parameters which characterize the *network information* available at a node taking part in stateless broadcasting.

2.1. Statement of objectives

We identify the salient features of stateless broadcasting which subsequently act as guidelines for our discussion in the rest of the paper. The performance objectives of any broadcast algorithm can be summarized as

$$\begin{aligned} \min \quad & n_{tx} \times e_{avg} \\ \text{s.t.} \quad & \sum_{i=1}^{N} r(i) = N \end{aligned} \tag{1}$$

where n_{tx} denotes the total number of broadcasts in the network while e_{avg} is the average energy consumption for a data transmission. The term e_{avg} is important as we use it as a measure of the energy consumption involved as part of the algorithm design. Thus, this accounts for additional overheads incurred by the broadcasting algorithm. The second part of (1) identifies the objective of network coverage. For a node i in a network with N nodes, $r(i)$ takes the value 1 when the node receives the broadcast and 0 otherwise.

The objectives outlined above can easily be mapped to the objective of addressing the *Broadcast Storm* problem. As with existing protocol designs and from the results in Bianchi (2000), it is safe to assume that minimizing the number of transmitting nodes (n_{tx}) results in minimizing the contention and collision overheads. A combination of both the conditions intuitively implies minimizing the redundancy while the parameter e_{avg} explicitly takes care of the energy costs.

The analysis methodology followed in this paper is based on modeling the fraction of forwarding nodes required for broadcasting. Subsequently, we use this model to obtain design insights from the point of view of ensuring reliability of broadcasting and the impact of protocol overheads (in terms of the resulting energy consumption).

2.2. Network information

We have earlier introduced *network information* as the information derived from overheard transmissions in the neighbourhood. In the context of stateless broadcasting, this needs to be mapped to the transmission effectiveness of the node in terms of the expected additional coverage (EAC). For distance based broadcasting schemes, the distance from the transmitting node is used to infer the EAC. In counter based schemes, the number of transmissions overheard is used as a measure. For protocol design using both schemes, a random or deterministic delay is introduced subsequent to a node receiving the first copy of the broadcast message. This is to increase the accuracy of the estimated EAC before it decides on its transmission behaviour. Using these salient features of stateless algorithms, we distinguish the *network information* available at a node into two distinct phases. As the delay used in both schemes is protocol specific, we do not focus on it as an aspect of *network information* in our analysis. Instead, we identify the effect of transmissions in a node's neighbourhood while it waits for channel access.

As part of recognizing the broadcast from a neighbouring node, the node can estimate the effectiveness of its transmission if it chooses to take part in forwarding. We term this as the *broadcast recognition phase* (*BRP*). Thus, the *network information* available can be expressed in terms of the distance to the transmitting node. We define a distance threshold denoted by a as a fraction of the transmission range. Given a uniform distribution of nodes, effectiveness of a node's transmission is maximized if it lies at the periphery of the transmission range. We consider a behaviour in which nodes receiving the first copy of message from a node within the range a choose not to forward. In our analysis, we examine the dependence of the forwarding probability of nodes as a function of a.

In addition to the *network information* obtained from the first phase, a node's transmission effectiveness also depends on the number of broadcasts in its neighbourhood, which forms a basis for counter based schemes (Ni et al., 1999). We refer to this as the *broadcast optimization phase* (*BOP*). Each overheard broadcast results in a reduction of the EAC at a node and therefore, the effectiveness of its transmission. Further, this dependence grows stronger for broadcasts received over shorter distances. Based on this, we consider a behaviour wherein nodes probabilistically decide on forwarding behaviour upon overhearing broadcasts from neighbours located nearer than a threshold distance, d_{th}. The packet discarding probability, p_{disc}, is also a function of the distance threshold, and is discussed in the next section.

Here, it is interesting to note how these parameters map to protocol specific behaviour of existing schemes. Firstly, existing mechanisms make use of a random delay to decide on their forwarding behaviour. Overheard broadcasts during this duration impact the node's forwarding behaviour which is determined at the end of the delay. By allowing the forwarding behaviour to be determined purely in terms of neighbourhood transmissions, we map the protocol specific operations to actual MAC layer ones. Further, the forwarding behaviour reduces to that of a basic distance based scheme when $d_{th} = a$ and $p_{disc} = 1$.

Defining the parameters based on the distance between nodes also allows us to stay true to the broadcast performance objective of identifying a CDS in the network. For the unit disk graph realization of a wireless network, construction of a CDS implies maximizing the distance between them (Wu et al., 2006). Further, maximizing the distance is also necessary for optimal scheduling among broadcasting nodes (Huang et al., 2007).

3. Broadcast performance analysis under ideal channel conditions

3.1. System model

We are motivated to understand the impact of *network information* at each of the two phases on the broadcast performance. It is necessary to map node transmission behaviour to the estimated effectiveness of the transmission based on the available *network information*. We define node transmission behaviour from the perspective of available *network information*. The network model we consider is that of a Unit Disk Graph (UDG) (Clark et al., 1990) in which all nodes have a fixed transmission radius of unit distance.

In the *broadcast recognition phase*, only nodes lying outside the range characterized by the distance threshold, a, take part in forwarding since all other transmissions are estimated to be ineffective. Similarly, in the *broadcast optimization phase*, the transmission effectiveness reduces probabilistically with the number of broadcasts heard over a close range. Therefore, we define a behaviour in which nodes probabilistically decide not to forward upon hearing a rebroadcast from a node lying closer than a threshold d_{th}. For ease of discussion, we use $d_{th}=a$. In order to ensure that the resulting behaviour closely maps to the effectiveness of the transmissions, we define p_{disc} on the basis of a set of observations. As mentioned earlier, the effectiveness of a transmission with respect to an overheard broadcast would depend on the distance between the two nodes. Specifically, a node's transmission is likely to be redundant if it is located very close to the broadcasting node. Thus, a smaller value d_{th} would imply that every overheard broadcast reduces the effectiveness at a faster rate. The converse would be true for larger values of d_{th}. The same would be true for different broadcasts received from within the range of d_{th}, given a fixed value for d_{th}. Averaging the effect of all transmissions from within the range of d_{th} and using $d_{th}=a$, we define p_{disc} as

$$p_{disc} = \left(1-\frac{a}{2}\right) \tag{2}$$

Considering a fixed unit transmission radius, we have $0<a<1$. To effectively understand the interplay between transmission behaviours of nodes, we consider the entire set of nodes as divided into regions of concentric circles marked by their transmission range, with the source at the centre. We term these regions as hop ranges. Thus, a node that requires at least one forwarding node, and therefore two transmissions, to reach the source lies in the second hop range. We refer to the transmission coverage region as *TR*. The region lying within the distance threshold is referred to as the alerted region and denoted by *AR*. Forwarding nodes are likely to be located in the region between *TR* and *AR*, which is denoted as *FR*. Since we use $d_{th}=a$, *AR* is also used to denote the range marked by the distance threshold for *broadcast optimization phase*. We use *MSG* to identify the packet being broadcast in the network.

3.2. Broadcast recognition phase

We first identify the effect of the available *network information* during the *broadcast recognition phase* on the forwarding probability of nodes. Since this does not include the effect of the *broadcast optimization phase*, the forwarding behaviour of nodes would only depend on the first copy of *MSG* received. Our discussion hereafter concerns the probability that a node in a hop range h either rebroadcasts an overheard packet or refrains from doing so. We denote these two probabilities by $p_t(h)$ and $p_r(h)$ respectively. We assume uniform distribution of nodes in the network.

We start by obtaining the transmission probabilities in the first hop range and subsequently obtain generalized expressions for the succeeding hop ranges. The analysis for $h=1$ is straightforward and is along the lines of analysis of existing distance based mechanisms (Williams et al., 2004; Chen et al., 2005). All nodes lying within the *AR* would refrain from forwarding while those lying in *FR* would transmit. Therefore, the forwarding probability values for the first hop range would be obtained as

$$p_t(1) = 1-a^2$$

$$p_r(1) = a^2 \tag{3}$$

Next we examine the transmission behaviour for nodes in $h>1$. Since the nodes are not within the transmission range of the source, the transmission behaviour of a node is determined by the distance to the node from which it receives the first copy of *MSG*. This can correspond to one of two different scenarios which can impact the behaviour. In the first case, a node in h overhears *MSG* from a node lying in $(h-1)$, which we term *Scenario* 1. Alternatively, a node in h could receive *MSG* from another node in h, which we call *Scenario* 2. By design, the broadcasting node in *Scenario* 2 would have heard *MSG* from a node in $(h-1)$ and lies in *FR*. A third scenario is that the transmission of RREQ from $(h+1)$ impacts transmission in h. However, the likelihood of this occurring is low since the node in h is likely to have overheard at least one of the preceding transmissions from $(h-1)$ and h (Fig. 1). Hence, we do not include this scenario in our derivations. As a result of the scenarios mentioned above, nodes that transmit and those that do not are unlikely to be distributed according to a specific geometric pattern for $h\geq 2$ unlike $h=1$. The analysis hereafter is focused on identifying the forwarding probability as a result of *Scenario* 1 and *Scenario* 2. These probability values are then used to obtain the fraction of nodes in the network that transmit.

As node broadcasting behaviour depends on the relative locations of the nodes with respect to each other, the overlapping regions between transmission ranges of nodes need to be considered, which can be obtained from circle geometry. The area of intersection between any two circles of radii r and R with the centres located at a distance d from each other can be given as

$$A(d,r,R) = r^2 \cos^{-1}\left(\frac{d^2+r^2-R^2}{2dr}\right) + R^2 \cos^{-1}\left(\frac{d^2+R^2-r^2}{2dR}\right) - \frac{1}{2}\sqrt{(-d+r+R)(d+r-R)(d-r+R)(d+r+R)} \tag{4}$$

Scenario 1: For any node in hop range h, the set of transmissions accounting for *Scenario* 1 lies in the overlapping area of its transmission range and region of $(h-1)$. The latter is the circle with radius $(h-1)$ (since transmission range is normalized to 1) with the source at its centre. The set of transmissions of *MSG* that constitutes *Scenario* 1 lie in the area of intersection of *AR* and the circular region for $(h-1)$. Thus, for any node located at a distance x

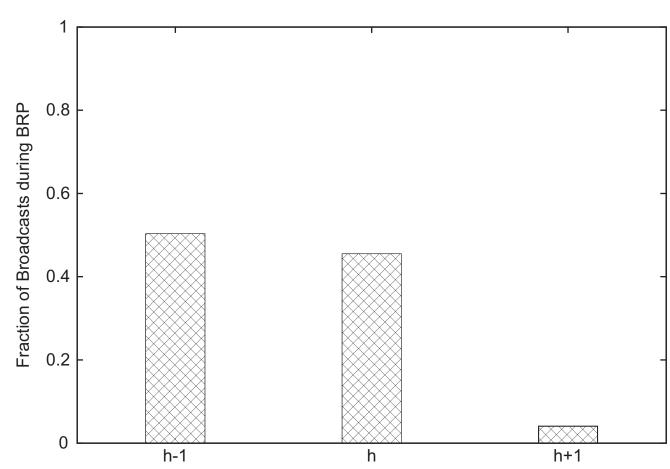

Fig. 1. Figure illustrating the fraction of broadcasts heard by a node in hop range h during *BRP*.

from the source and lying in the hop range h, the probability that the copy *MSG* it receives first is from the hop range $(h-1)$ and it refrains from broadcasting the overheard MSG can be given as

$$p_r(h,h-1,x) = p_t(h-1)\frac{A(x,a,h-1)}{A(x,1,h-1)} \tag{5}$$

It is important to clarify our use of the notation here, so as to understand the difference between $p_r(h)$ and $p_r(h,h',x)$. The latter notation $p_r(h,h',x)$, as used above, is the probability that any node, located at a certain distance x from the source, refrains from broadcasting upon overhearing the first copy of *MSG* from a node in h'. This will be used later to obtain the probability that any node in hop range h refrains from transmission, given as $p_r(h)$.

Scenario 2 (*S*2): Similar to *Scenario* 1, the probability that a node decides not to forward *MSG* in this case would depend on the size of the overlapping region between the hop range h and the *AR* for the node. However, this probability would also indirectly depend on the transmissions taking place in $(h-1)$. We term this as *Phenomenon-S*2 and elaborate on this here. If we visualize the way *MSG* is propagated across the network, the first copy to enter hop range h must have been transmitted by some node in $(h-1)$. Subsequently, the set of nodes in h, say T_h, which are located in the *FR*s of the transmitted copies of *MSG* contend to broadcast their copies. If one of the nodes in T_h transmit in the succeeding time slot (i.e. immediately after having received *MSG* from $(h-1)$), the set of nodes whose transmission behaviour it impacts would only be those which are not in the range of the previous transmission of *MSG* (from $(h-1)$). The nodes impacted by a later transmission would similarly be those that are outside the range of all the preceding transmissions. Here, it is important to note that the set of nodes that T_h contends with include those in $(h-1)$ that are yet to transmit. Thus, the probability that the first copy of *MSG* a node in h receives, is from a node in h, is the probability that it has not overheard any of the preceding broadcasts from $(h-1)$.

To better understand *Phenomenon-S*2, we observe Fig. 2 in which nodes A and B are located in the second hop range while nodes C and D are in the third, S being the source. Subsequent to the source having transmitted *MSG*, we fast forward to the state in which all nodes in the second hop range have received *MSG*. We assume that both A and B take part in broadcasting on account of being located in *FR* when receiving their first copies of *MSG*. Suppose A first gains access to the channel with only C overhearing its transmission in the third hop range. Since C does not lie within A's *AR*, it attempts to rebroadcast and contends for channel access with B, which has yet to transmit. Therefore, the probability that D's transmission behaviour is determined by C depends on the probability that the latter transmits before B does.

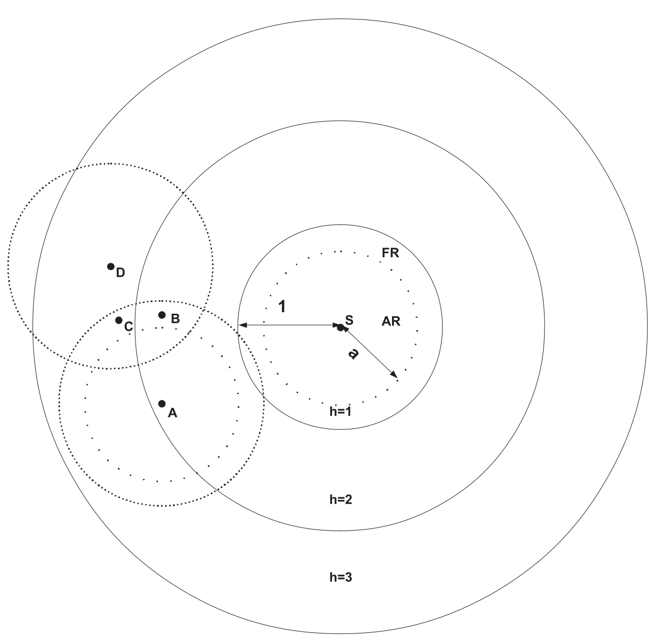

Fig. 2. Figure illustrating the distribution of forwarding nodes.

Thus, the analytical model for *Scenario* 2 needs to include *Phenomenon-S*2. As discussed above, this is the probability that a node i in h overhears *MSG* from another node in h before one from $(h-1)$. This would depend on how channel contention proceeds among i's neighbours. Now, the channel access probability for any node would depend on the size of this neighbourhood. From Fig. 2, we can conclude that the transmission probability of a node in $(h-1)$ would reduce as more nodes in h start contending upon having received a copy of *MSG*. It is easy to see that the number of nodes in h which start contending would depend on the value of a since only those which lie outside *AR* would enqueue their received copies for transmission. While it is difficult to geometrically estimate the contending probability of nodes in this scenario, based on the above reasoning, we can conclude that the transmission probability of a node in $(h-1)$ scales with $(1-a^2)$, on average. Thus, we can approximate the probability that a transmission from $(h-1)$ does not reduce the effectiveness of one in h as

$$p_c = [1-(1-a^2)p_t(h-1)] \tag{6}$$

It is worth noting here that a similar phenomenon is likely to take place for (5) where transmission from the hop range h could influence the effectiveness of transmissions from $(h-1)$. However, the probability of the same is expected to be relatively small since any node in h would have received *MSG* either directly or indirectly from $(h-1)$. For any other node still contending in $(h-1)$, the additional coverage area in h would have been reduced by other transmissions from $(h-1)$. Hence, the effect of a subsequent transmission from h would be negligible.

The probability that a node in h refrains from forwarding in *Scenario* 2 follows from the above discussion as

$$p_r(h,h,x) = [1-(1-a^2)p_t(h-1)]p_t(h)\frac{(A(x,a,h)-A(x,a,h-1))}{(A(x,1,h)-A(x,1,h-1))} \tag{7}$$

The probability that a node refrains from forwarding is the probability that it is located in *AR* when receiving the first copy of *MSG*, in either *Scenario* 1 or *Scenario* 2. Now, a node situated at a distance x from the source can lie anywhere on the circle with circumference $2\pi x$ while the region covered in the hop range h is a concentric circle with area $(2h-1)\pi$. Thus, the probability that any node in the hop range h refrains from forwarding *MSG* can be expressed as

$$\begin{aligned} p_r(h) &= \int_{h-1}^{h} p_r(h,h-1,x)\frac{2x}{2h-1}\,dx + \int_{h-1}^{h} p_r(h,h,x)\frac{2x}{2h-1}\,dx \\ &= \int_{h-1}^{h} p_t(h-1)\frac{A(x,a,h-1)}{A(x,1,h-1)}\frac{2x}{2h-1}\,dx \\ &\quad + \int_{h-1}^{h}[1-(1-a^2)p_t(h-1)]p_t(h) \\ &\quad \times \left(\frac{A(x,a,h)-A(x,a,h-1)}{A(x,1,h)-A(x,1,h-1)}\right)\left(\frac{2x}{2h-1}\right)dx \end{aligned} \tag{8}$$

The probability that any node in hop range h transmits is that it does not refrain from doing so and can be obtained as $p_t(h) = 1-p_r(h)$. Solving the implicit linear equation, $p_t(h)$ can be expressed in terms of the transmission probability of the previous hop range as

$$p_t(h) = \left(\frac{1 - \int_{h-1}^{h} p_t(h-1)\frac{A(x,a,h-1)}{A(x,1,h-1)}\frac{2x}{2h-1}dx}{1 + \int_{h-1}^{h}[1-(1-a^2)p_t(h-1)]\frac{(A(x,a,h)-A(x,a,h-1))}{(A(x,1,h)-A(x,1,h-1))}\frac{2x}{2h-1}dx} \right) \tag{9}$$

Using (3) in (9) and subsequently in (8), we obtain the probability values for the second hop range and in turn use the values for succeeding hop ranges.

Given the node transmission probabilities for successive hop ranges, the fraction of all nodes that take part in route discovery can be obtained by summing the fraction of nodes transmitting in each hop range. As discussed before, the first hop range is the circular area covered by the transmission range of the source while each subsequent hop range is a concentric circle with increasing area. The fraction of transmitting nodes can thereby be obtained as

$$F_{tx} = \frac{p_t(1)\pi + \sum_{h=2}^{H} p_t(h)(2h-1)\pi}{\pi H^2} \tag{10}$$

where H is the maximum number of hop ranges which would depend on the network topology. In our simulation scenario, the source s is located at the centre while the destination d is located at the edge of the circular simulation region. Therefore, the value of H can be obtained as $H = dist(s,d)/TX\_RANGE$, where $dist(s,d)$ is the distance between the source and the destination while TX_RANGE is the constant transmission range for any node in the network. The area of the total simulation region is, therefore, πH^2.

3.3. Broadcast optimization phase

Next, we evaluate the additional performance benefits achievable by utilizing *network information* available during the *broadcast optimization phase*. The resulting forwarding probability of nodes would depend on the discarding probability p_{disc} in addition to the value of a. Here, we observe that any node that can utilize the *network information* available in the *broadcast optimization phase* is one of the potential forwarding nodes after the *broadcast recognition phase*. For any particular hop range h, this is the same as the transmission probability obtained in (9), given as $p_t(h)$. Thus, the probability that a node refrains from forwarding as a result of the *network information* available in the *broadcast optimization phase* follows from (8) as

$$p_{rd}(h) = p_{disc}p_{ts}(h)\left[\int_{h-1}^{h} p_t(h-1)\frac{A(x,a,h-1)}{A(x,1,h-1)}\frac{2x}{2h-1}\,dx + \int_{h-1}^{h} p_{ts}(h)\frac{(A(x,a,h)-A(x,a,h-1))}{(A(x,1,h)-A(x,1,h-1))}\frac{2x}{2h-1}\,dx\right]. \tag{11}$$

The probability that any node refrains from forwarding after both the phases can, therefore, be expressed as

$$p_{rs}(h) = p_r(h) + p_{rd}(h) \tag{12}$$

As earlier, the corresponding transmission probability can be obtained as $p_{ts}(h) = 1 - p_{rs}(h)$. The fraction of nodes that transmits in the entire network can be obtained by substituting $p_{ts}(h)$ for $p_t(h)$ in (10).

3.4. Effect on redundancy

Having obtained the forwarding probability of nodes in stateless broadcasting, we are interested in analyzing the effect on the redundancy of transmissions. As discussed before, a node's transmission probability needs to be closely mapped to its effectiveness, which is in terms of the EAC (Ni et al., 1999).

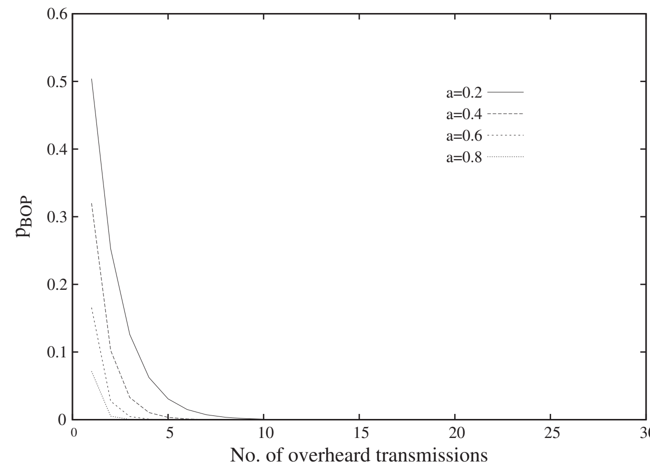

Fig. 3. Figure showing probabilities of retransmission.

For the *broadcast recognition phase*, the effect of *network information* on redundancy is straightforward. Since node transmission probabilities only depend on the first copy of *MSG* received, the effect is deterministic. The only reduction in redundancy is due to the fraction of nodes that does not transmit on account of having being located in the *AR* upon receiving the first copy of *MSG*.

In the case of the *broadcast optimization phase*, we identify how (2) translates into reducing the number of redundant transmissions. We derive the probability of a node discarding its packet upon overhearing multiple broadcasts of *MSG*. As observed in Ni et al. (1999), the probability should reduce exponentially in order to minimize redundancy. We identify the corresponding probability for any node within the range of the source. Since we analyze the effects on redundancy as a result of the *broadcast optimization phase*, the nodes under consideration lie in the *FR*. We denote by p_{AR} and p_{FR} the probabilities that a node lies in *AR* or *FR* respectively for any subsequent broadcast of *MSG* that it overhears. The probability that a node transmits upon hearing a broadcast from within *AR* can be given as $p_{AR}(1-p_{disc})$. Hence, the probability that a node transmits after having overheard n broadcasts in the *broadcast optimization phase* (*BOP*) can be given as

$$p_{BOP}(n) = \sum_{k=0}^{n} \binom{n}{k} [p_{AR}(1-p_{disc})]^k p_{FR}^{(n-k)} \tag{13}$$

The corresponding values are plotted in Fig. 3. Note that the value of n does not include the first copy of *MSG* received as part of the *broadcast recognition phase*. As the value of a[1] increases, the forwarding probability of a node drops sharply with the number of overheard broadcasts, which implies fewer redundant transmissions. A higher value of a implies that a greater set of nodes is located within the *AR* for a rebroadcast copy of *MSG*. From the figure, we can conclude that a stricter control over redundancy is exercised with increasing values of a; with $a = 0.6$ being the closest to approximating the effectiveness of transmissions based on Ni et al. (1999).

3.5. Simulation results

We validate the analytical results on the transmission probability of nodes in the context of a broadcasting application.

[1] As before, $a < 1$ since the transmission range is normalized to 1.

Evaluating broadcasting performance in the context of an application allows us to examine its impact on other performance metrics as well. The broadcasting application we consider here is route discovery for a reactive routing protocol in which nodes decide on whether or not to forward the route request based on *network information* obtained from the *broadcast recognition* and *broadcast optimization phases*. We use NS-2 (Network Simulator) to simulate the routing performance.

The route discovery mechanism is based on the AODV routing protocol (Perkins et al., 2003). Route discovery is initiated by a source node in the absence of an existing route to the destination by broadcasting a route request (RREQ) after which it waits for the route reply (RREP). In our simulations, the RREQ is propagated throughout the network based on the estimated distance to packets received in the *BRP* and *BOP*. We obtain results for different values of a, with $a=0$ implying pure flooding. We consider an IEEE 802.11b network. Nodes contend for channel access using the distributed coordination function (DCF).

Since our analysis focuses on the performance benefits achievable from *network information* at different stages of broadcasting, it is difficult to compare the proposed model with existing models that consider individual protocol specific parameters such as the delay (Williams et al., 2004). Nevertheless, parallels can still be drawn between the information available during the *BRP* and the analytical model proposed for distance based broadcasting in Williams et al. (2004). For the *BOP*, however, the relation is less straightforward due to the fact that our analysis considers the entire *network information* available during the *BOP* in terms of the distance and the effect of multiple transmissions. Existing counter based strategies, on the other hand, only consider the number of transmissions overheard. To illustrate the accuracy of our proposed model, therefore, we make a qualified comparison by comparing the results for *BRP* with that of the model for distance based scheme proposed in Williams et al. (2004).

The simulation results for the fraction of forwarding nodes are shown in Fig. 4. The simulation results are shown to match quite closely with the proposed analytical model. The accuracy of the proposed model is particularly highlighted in Fig. 4(a) when compared to the model proposed by the authors in Williams et al. (2004). One reason for this is that, as part of our analysis, we do not assume uniform transmission probabilities in the neighbourhood of a node. As our analysis is recursive over successive hop-ranges, the transmission probability of a node is obtained as a function of the locations of the nodes in its neighbourhood. Further, we also scale the transmission probability with the contention among transmitting nodes.

To put in perspective how the performance benefits achievable using *network information* compare with existing algorithms, we plot the performance of the DIS_RAD algorithm proposed in Chen et al. (2005). The authors in Chen et al. (2005) propose a design which incorporates features of distance based algorithms in counter based broadcasting with the specification of separate random access delays (RADs) depending on whether a node lies within a distance threshold or not. Such a design shares, to an extent, our motivation of utilizing the *network information* available at a node. Note, however, that the performance of the algorithm critically depends on predetermined values of protocol specific parameters of the RAD and the counter threshold (C_{Th}) which do not take into consideration the instantaneous information available at the nodes. Fig. 4(b) compares the analytical performance of DIS_RAD with $C_{Th}=5,6$ (ensuring 100% coverage) to the performance achievable with *network information* available during *BOP*, as obtained from our analysis. The DIS_RAD algorithm results in greater than 70% nodes acting as forwarding nodes. However, a performance of less than 40% can be achieved if instantaneously available *network information* is utilized by nodes, as obtained using our model.

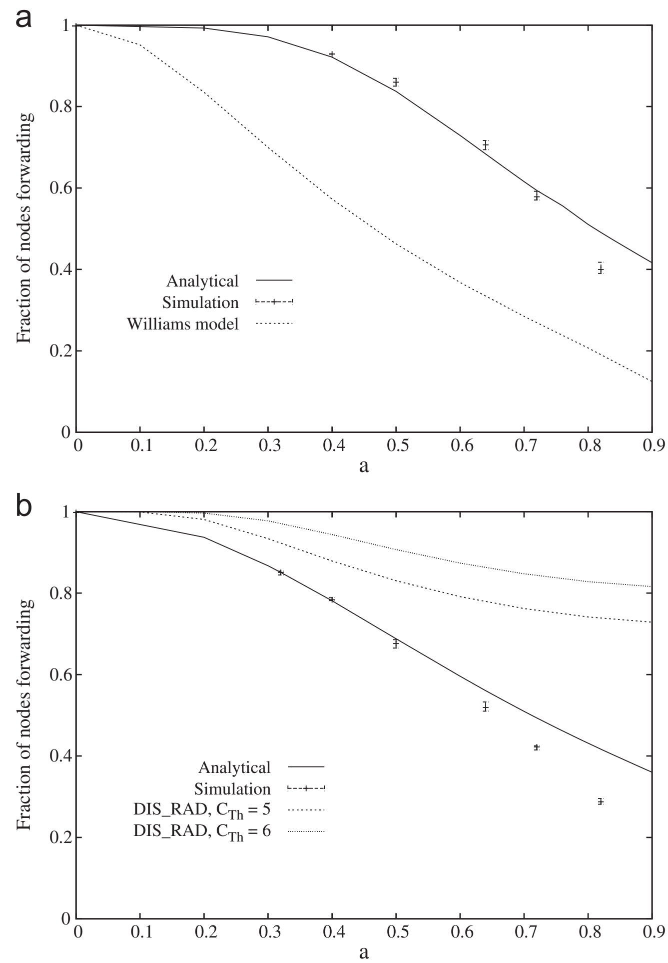

Fig. 4. Fraction of forwarding nodes when *network information* from (a) only *BRP* is used, (b) both *BRP* and *BOP* is used.

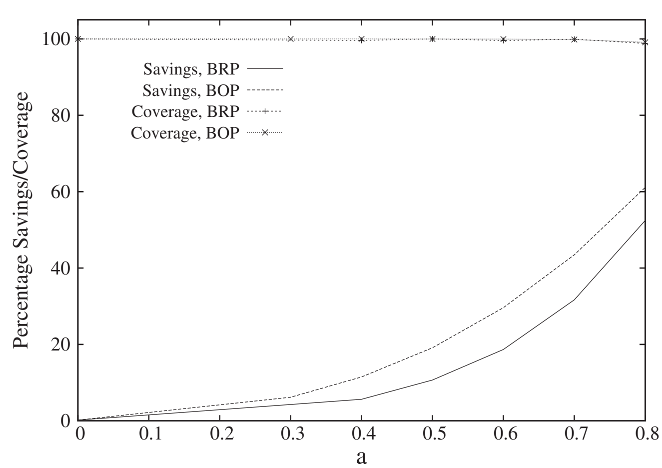

Fig. 5. Savings achieved at the *broadcast recognition* and *broadcast optimization phases* and the corresponding network coverage.

While transmission probability is a crucial aspect of the broadcasting performance, the final performance of the broadcasting application, route discovery in this case, needs to be evaluated in terms of the relevant metrics. We evaluate the route discovery performance in terms of the savings, resulting hop count and the time taken for route discovery. We obtain results for the performance benefits corresponding to varying values of a with $a=0$ corresponding to that of the AODV routing protocol.

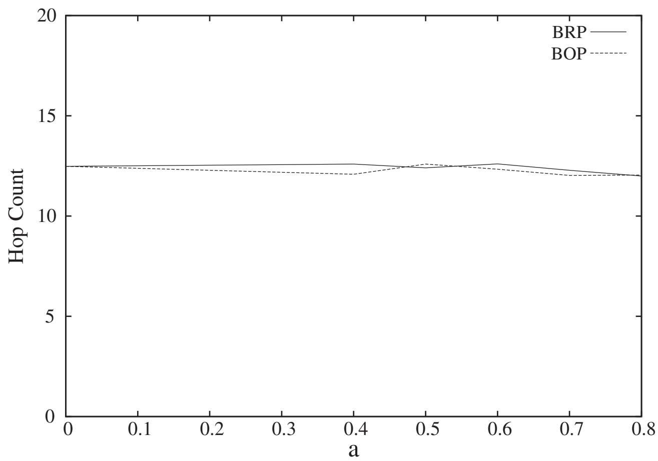

Fig. 6. Hopcount distribution.

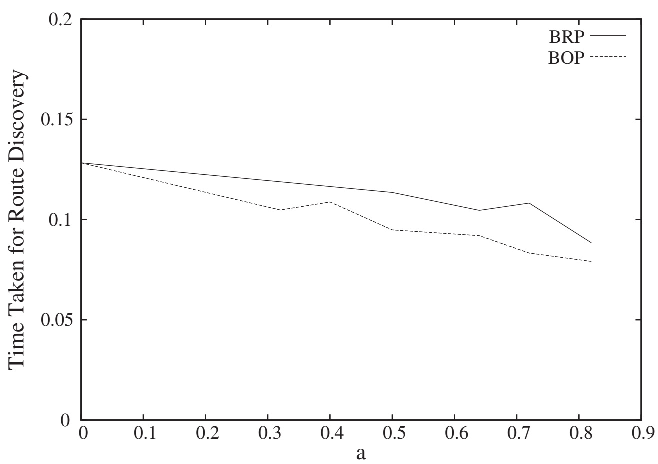

Fig. 7. Route discovery delay.

As in Ni et al. (1999), the savings is obtained as the Savings Rebroadcast Ratio (SRB) which is given as $SRB=(r-t)/r$, where r is the number of nodes receiving the broadcast message while t is the number of them that actually transmit. Fig. 5 shows the percentage savings obtained in terms of the number of transmissions as well as the corresponding network coverage. We see that the network coverage does not suffer upon increasing the value of a. Finally, we observe the effects of reduced number of transmissions on the performance of the routing protocol in terms of hop count. As shown in Fig. 6, the hop count does not vary much upon varying the value of a. Thus, use of stateless broadcasting algorithms does not degrade the routing performance. Further, the effect of utilizing *network information* reduces the time taken for route discovery, as shown in Fig. 7. This is expected since fewer nodes are involved in broadcasting with increasing values of a, leading to lesser contention and collision overheads.

4. Insights for algorithm design

In this section, we discuss the implications of our analysis for design of effective broadcast algorithms. We use our results to obtain insights on the feasibility conditions in terms of the broadcast reliability given the network density and the tradeoff between broadcast performance and network costs. The first part of the discussion here looks at optimal parameter choice for stateless algorithm design to ensure reliability. The second part identifies feasibility conditions for design of stateful algorithms in terms of the permissible network costs that can be incurred for acquiring state information.

4.1. Reliability of broadcasting

The nature of stateless broadcasting implies that it can be used as the benchmark for broadcasting performance in a given network. However, as the performance analysis in the previous section hinges on the choice of parameters for stateless broadcasting, it is necessary to understand how the parameter values depend on the network characteristics. We evaluate how the node density in a network impacts the choice of the distance threshold a so as to ensure reliability of broadcasting and therefore limits the achievable performance.

To understand the effect of a on reliability, we focus on the choice of a based on the node density so as to ensure that the message *MSG* is propagated. The latter would fail to happen if, at any stage, the transmission of the message from a node is not propagated beyond the immediate one hop neighbourhood. We define *transfer probability* as the probability that a message transmitted by a node is received by at least one node in its second hop-range. In other words, this is the probability that the message has been propagated beyond the current neighbourhood.

Upon being transmitted by either the source or any other node, the message *MSG* is retransmitted by other nodes in the neighbourhood depending on the transmission probability as obtained earlier. The probability that any such transmission from a node located at a distance x is received by a node in the second hop-range can be obtained as $(\pi-A(x,1,1))/3\pi$. The probability that any transmission from the first hop-range is received by a node in the second hop range is, therefore, $p_{th}=\int_0^1((\pi-A(x,1,1))/3\pi)2x\,dx$. Given a node density ρ, the number of nodes in the first hop-range is $n_1=\pi\rho$ and in the second hop-range $n_2=3\pi\rho$. Thus, the probability that a transmission made in the first hop-range is not received by any node in the second one is

$$p_{2nr}=(1-p_{th})^{n_2}. \tag{14}$$

The scenario that the message is not propagated to the second hop-range, thereby resulting in it dying, can result due to either of two conditions. Firstly, this could result if none of the neighbours in the first hop-range transmits as a result of the choice of a. Denoting the transmission probability as p_{tx}, the probability that none of the first hop neighbours transmits can be obtained as $(1-p_{tx})^{n_1}$. Subsequently, given that at least one node transmits, the message propagation may still be halted if none of the transmissions is received by any node in the second hop-range. The probability of the occurrence of this event is $\sum_{k=1}^{n_1}p_{2nr}^k\binom{n_1}{k}p_{tx}^k(1-p_{tx})^{(n_1-k)}$. Thus, the total probability that the message transmitted from a node is not propagated to the second hop-range is

$$p_{nr}=(1-p_{tx})^{n_1}+[1-(1-p_{tx})^{n_1}]\sum_{k=1}^{n_1}p_{2nr}^k\binom{n_1}{k}p_{tx}^k(1-p_{tx})^{(n_1-k)} \tag{15}$$

The *transfer probability* is thereafter obtained as

$$p_{tf}=1-p_{nr} \tag{16}$$

The value of p_{tf} obtained above expresses the impact of the choice of a on the broadcasting reliability. Thus, the target broadcasting performance for a given node density would correspond to the highest value of a for which $p_{tf}=1$. We obtain numerical results for the transmission from the source for increasing node densities. The corresponding *transfer probabilities* for both the *BRP* and *BOP* are shown in Fig. 8. Thus, while it is attractive to have an aggressive broadcasting strategy that minimizes the number of transmitting nodes, which corresponds to a

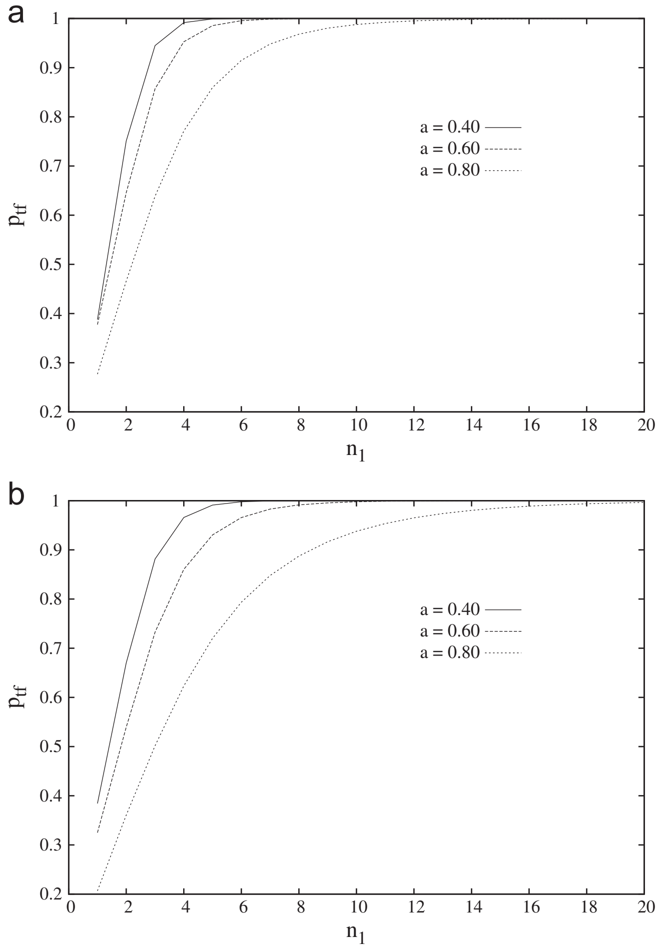

Fig. 8. Transfer probabilities over varying node densities. (a) Broadcast recognition phase (BRP), (b) broadcast optimization phase (BOP).

high value of a, this is only feasible for very high node densities. Thus, a broadcasting strategy which uses $a = 0.80$ that can achieve up to 60% savings is only feasible when the node density is such that the average number of first hop neighbours is at least 12 and 18 nodes for *BRP* and *BOP* respectively.

4.2. Network costs versus broadcast performance

Since stateless broadcasting operations do not require any prior information at nodes, an obvious motivation is to investigate the additional benefits achievable if some information were indeed made available. Acquiring such information would require additional transmission overheads, which would need to be offset by the broadcasting performance gains. To illustrate this tradeoff, we analyze the effect on stateless broadcasting if the *network information* used earlier were to be itself transmitted explicitly instead of being available as part of the overheard transmissions. An intuitive measure of network costs is the energy consumed by the network and hence, we evaluate the tradeoffs involved in energy consumption.

In Banerjee et al. (2010), we propose a stateless route discovery mechanism for reactive routing protocols such as AODV. As part of this, the AODV Route Request, AODV_RREQ is accompanied by an additional packet, AODV_RREQ_ALERT. This additional packet is transmitted over a shorter range and consumes less transmission power. All nodes that receive an AODV_RREQ_ALERT along with the AODV_RREQ do not forward the route request while the rest do. Based on this, we consider a stateless broadcasting algorithm in which nodes transmit an additional message *ALERT* over a range given by a when forwarding the message *MSG*. A node that receives the *ALERT* in the *broadcast recognition phase* (*BRP*) chooses not to take part in forwarding. A node that receives an *ALERT* during the *broadcast optimization phase* (*BOP*) chooses not to forward with a probability p_{disc}. We examine how transmission of additional *ALERT* messages impacts the energy consumption of the network.

We use existing energy models to determine the energy consumption of the network (Wieselthier et al., 2002). The minimum transmission power required by a node to ensure that the transmitted signal is received correctly at a distance r can be given as $E = t_{rx} r^{\alpha}$ where α depends on the communication medium and typically takes values between 2 and 4 and t_{rx} is the reception threshold. Normalizing the transmission range to 1 as before, the power consumed at a node due to forwarding a copy of *MSG* would be $E_{MSG} = t_{rx}$ and that of an ALERT, $E_{ALERT} = t_{rx} a^{\alpha}$. In the case of pure flooding which incurs no additional overhead, the energy consumed in the network would be the total energy consumption of all transmissions. The total energy consumed would, therefore, be $E_{Tot} = n_f t_{rx}$ where n_f is the number of transmissions. In the case where *ALERT*s are used, they contribute to the total energy consumption for the network in addition to the forwarded copies of *MSG*. Thus, if $n_a = f_a n_f$ denotes the number of transmissions in the network in such a scenario, the total energy can be given as $E_{aTot} = n_a t_{rx}(1 + a^{\alpha})$, where $f_a \le 1$. Now, for the scenario with *ALERT*s to be feasible, the condition $E_{aTot} < E_{Tot}$ would need to be satisfied, which can be rewritten as

$$n_a t_{rx}(1 + a^{\alpha}) < n_f t_{rx} \tag{17}$$

Using $f_a = n_a / n_f$, we arrive at the feasibility condition for which the mechanism can be implemented, that is

$$f_a < \frac{1}{1 + a^{\alpha}} \tag{18}$$

This implies that the fraction of nodes that transmit upon implementation of the broadcast algorithm cannot exceed $1/(1 + a^{\alpha})$. Similarly, any stateful broadcasting mechanism can be expected to have a feasibility condition of a similar form if the overheads involved in acquiring the network state can be expressed in similar terms as above. As an example, consider stateful protocols where additional neighbourhood information is obtained by exchanging "HELLO" packets. Without going into the specifics of such a protocol, we can identify the additional costs involved as the number of "HELLO" packets transmitted per node. Thus, in this case, if n_h is the number of "HELLO" transmissions

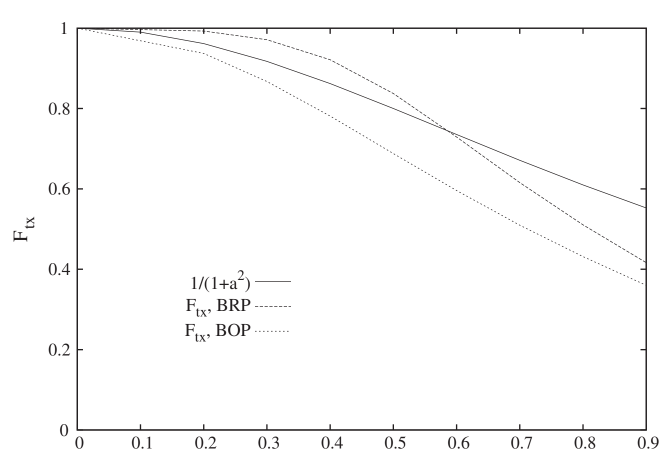

Fig. 9. Figure illustrating the relation between broadcasting performance and feasibility condition.

and n_{tx} is the number of broadcasts resulting from the protocol design, we would have $E_{aTot}=(n_h+n_{tx})t_{rx}$. Plugging its value into the first part of (17), one can derive the feasibility condition. Alternatively, if one were to map directly to the condition in (18), the number of "HELLO" messages could be imagined as being transmitted by all the n_{tx} broadcasting nodes. The number of "HELLO" transmissions per broadcasting node would then be n_h/n_{tx} which can be equated to a.

In our case, the energy consumed by the broadcasting mechanism using *ALERT* messages can be identified in terms of the values F_{tx} obtained for *BRP* and *BOP* in Sections 3.2 and 3.3 respectively. Fig. 9 compares these values to the R.H.S. of (18). We set the value of α to 2, which is most commonly used in the existing literature. We observe that, if the information obtained from *ALERT*s is only used in the *BRP*, the algorithm is only feasible if $a>0.6$ although if it is also used in the *BOP*, it is feasible for all values of a.

5. Stateless broadcasting for dynamic channel conditions

For our analysis so far, we consider the Unit Disk Graph (UDG), which is popular in wireless networking research due to its simplicity. In our case, such a model allows us to study the impact of design parameters on the broadcasting performance under idealized channel conditions. However, the assumption of a fixed transmission range is unlikely to be true in actual wireless network deployment due to dynamically varying channel characteristics. Hence, the performance of broadcasting algorithms also need to be understood in the context of dynamic channel conditions. In this section, we study the impact of unstable transmission ranges on broadcasting performance and how the parameter choice can be optimized in such a scenario.

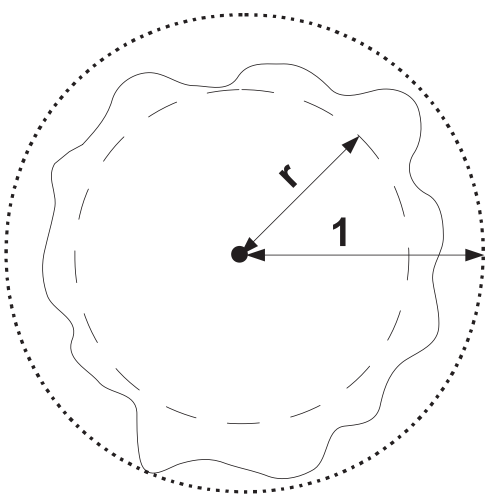

Fig. 10. Quasi-unit disk model with $0<r\leq 1$.

5.1. System model

A closer approximation to realistic channel conditions than the UDG model is provided by the Quasi-Unit Disk Graph (QUDG) (Kuhn et al., 2003). Two nodes in a QUDG are connected if the distance between them is less than r, $0<r\leq 1$ and disconnected if the distance is greater than 1. Nodes may or may not be connected if the distance separating them is between r and 1. The illustration of a Quasi-Unit Disk model is shown in Fig. 10. For any pair of such nodes separated by a distance x, $r<x\leq 1$, there exists an edge with probability $(1-x)/(1-r)$ (Bruck et al., 2009).

We use the QUDG model to evaluate the effect of the dynamicity of transmission ranges on the broadcasting performance. We show analytically how the value of r impacts the reliability of broadcasting for different values of a.

5.2. Analysis

For a realistic network modeled as a Quasi-Unit Disk Graph, the performance of stateless broadcasting hinges on the values of both the distance threshold a as well as r. To understand this, we refer to Fig. 11. In the first part of the figure, we have $r>a$. This implies that in the *broadcast recognition phase* (*BRP*), the only nodes that are certain to transmit are those located in the region $(a,r]$ while those in the region $(r,1]$ only transmit if they receive the original transmission. In the second part of Fig. 11, $a>r$ implies that the message *MSG* is propagated by those nodes located in $(a,1]$ that receive the message, thereby increasing the uncertainty. We analyze the reliability of broadcasting resulting from these two factors. Our analysis here is focused on the *BRP* upon which we later obtain insights for the *broadcast optimization phase* (*BOP*).

We first obtain the transmission probabilities for the $r>a$ case. In the first hop-range, the probability that any node is located in the region $(a,r]$ is (r^2-a^2), the transmission probability for which is 1. For any node located at a distance x from the source, $x\in(r,1]$, the probability of reception of the transmitted message is $(1-x)/(1-r)$. Thus, the total probability that any node in the first hop-range broadcasts the message transmitted by the source is

$$p_t^Q(1)=(r^2-a^2)+\int_r^1\frac{1-x}{1-r}2x\,dx=(r^2-a^2)+\frac{1+r-2r^2}{3} \tag{19}$$

The transmission probabilities for $h>1$ are obtained in the same manner as before. We use the same notations as earlier. For a node i located in the hop-range h at a distance x from the source,

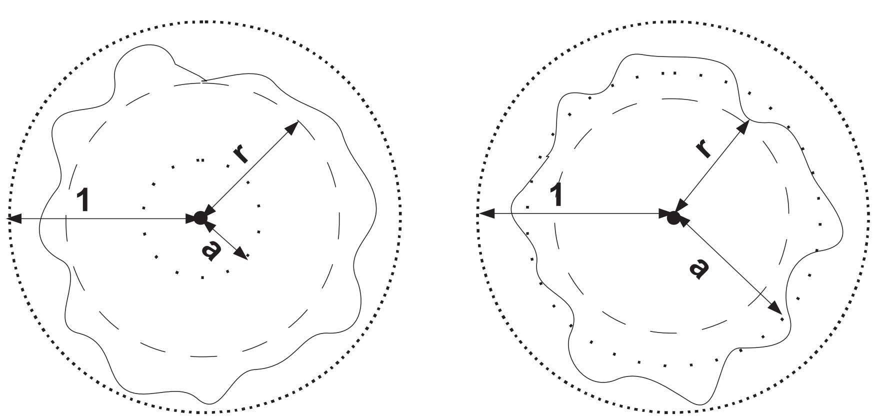

Fig. 11. Quasi-unit disk model illustrating $a<r$ and $a>r$.

the set of nodes in the hop-range $(h-1)$ from which it can receive the broadcast message are located in the region $A(x,1,h-1)$. Among the nodes located in this region, all transmissions from nodes in the region $A(x,r,h-1)$ can be received without error. However, transmissions from nodes located in the region $A(x,1,h-1)-A(x,r,h-1)$ would be received with a probability $(1+r-2r^2)/3$. The probability that i refrains from broadcasting upon receiving is that it has received the message from a node located in the region $A(x,a,h-1)$,

$$p_{rf}^{Q}(h,h-1,x) = p_t^{Q}(h-1) \times \frac{A(x,a,h-1)}{A(x,r,h-1)+\frac{1+r-2r^2}{3}[A(x,1,h-1)-A(x,r,h-1)]} \tag{20}$$

Similarly, the total area in the hop-range h from which MSG can be received can be obtained as $A'(x,h,h) = [A(x,r,h)-A(x,r,h-1)]+((1+r-2r^2)/3)[(A(x,1,h)-A(x,1,h-1))-(A(x,r,h)-A(x,r,h-1))]$. Subsequently, the probability that node i refrains from broadcasting upon receiving a transmission from the hop-range h can be obtained similar to (7) as

$$p_{rf}^{Q}(h,h,x) = [1-(1-a^2)p_t^{Q}(h-1)]p_t^{Q}(h)\frac{A(x,a,h)-A(x,a,h-1)}{A'(x,h,h)} \tag{21}$$

Thus, the total probability that a node in the hop-range h refrains from rebroadcasting the message MSG is

$$p_{rf}^{Q}(h) = \int_{h-1}^{h} p_{rf}^{Q}(h,h-1,x)\frac{2x}{2h-1} + \int_{h-1}^{h} p_{rf}^{Q}(h,h,x) \tag{22}$$

The probability that a node in hop-range h transmits after having received a copy of MSG can be obtained by substituting $p_t^{Q}(h) = 1-p_{rf}^{Q}(h)$ which is similar to (9).

For the case $a>r$, we notice that in the first hop-range, the probability that a node retransmits the received broadcast is the probability that it is located at a distance x, where $a<x<1$ and it also receives the transmitted message. Thus, the transmission probability of a node located in the first hop-range is

$$p_t^{Q}(1) = \int_a^1 \frac{1-x}{1-r}2x\,dx = \frac{(1-a)(1+a-2a^2)}{3(1-r)} \tag{23}$$

For a node in the hop-range h located at a distance x from the source, the probabilities $p_{rf}^{Q}(h,h-1,x)$ and $p_{rf}^{Q}(h,h,x)$ are obtained as

$$p_{rf}^{Q}(h,h-1,x) = p_t^{Q}(h-1) \times \frac{A(x,r,h-1)+\frac{1+r-2r^2}{3}[A(x,a,h-1)-A(x,r,h-1)]}{A(x,r,h-1)+\frac{1+r-2r^2}{3}[A(x,1,h-1)-A(x,r,h-1)]}$$

$$p_{rf}^{Q}(h,h,x) = [1-(1-a^2)p_t^{Q}(h-1)]p_t^{Q}(h)\frac{A''(x,h,h)}{A'(x,h,h)} \tag{24}$$

where $A''(x,h,h) = [A(x,r,h)-A(x,r,h-1)]+((1+r-2r^2)/3)[(A(x,a,h)-A(x,a,h-1))-(A(x,r,h)-A(x,r,h-1))]$.

Subsequently, the values of $p_{rf}^{Q}(h)$ and $p_t^{Q}(h)$ are obtained in the same manner as for $r>a$.

Having obtained the node transmission probability, we now analyze how the reliability of broadcasting is affected as a result of variable transmission ranges. We obtain the probability p_{nr} that a node does not receive a copy of the message MSG even if all nodes located in its first hop-range receive it, i.e. all of them refrain from transmitting. Note that the *transfer probability* defined earlier in Section 4.1 can also be used as a measure of the reliability of broadcasting in terms of the probability that the message is propagated by the source. We choose p_{nr} instead to demonstrate the probability of the broadcasting process breaking down irrespective of whether it is propagated by the source.

In the QUD model, the uncertainty of reception of transmissions in the range $(r,1]$ implies that a node can only receive a fraction of all possible transmissions in its neighbourhood. Thus, given a node density ρ, the total number of nodes in the first hop range is $n=\pi\rho$. Of these, the expected number of transmissions $n(r)$ that a node can receive depends on the value of r,

$$n(r) = \frac{(1+r+r^2)}{3}n \tag{25}$$

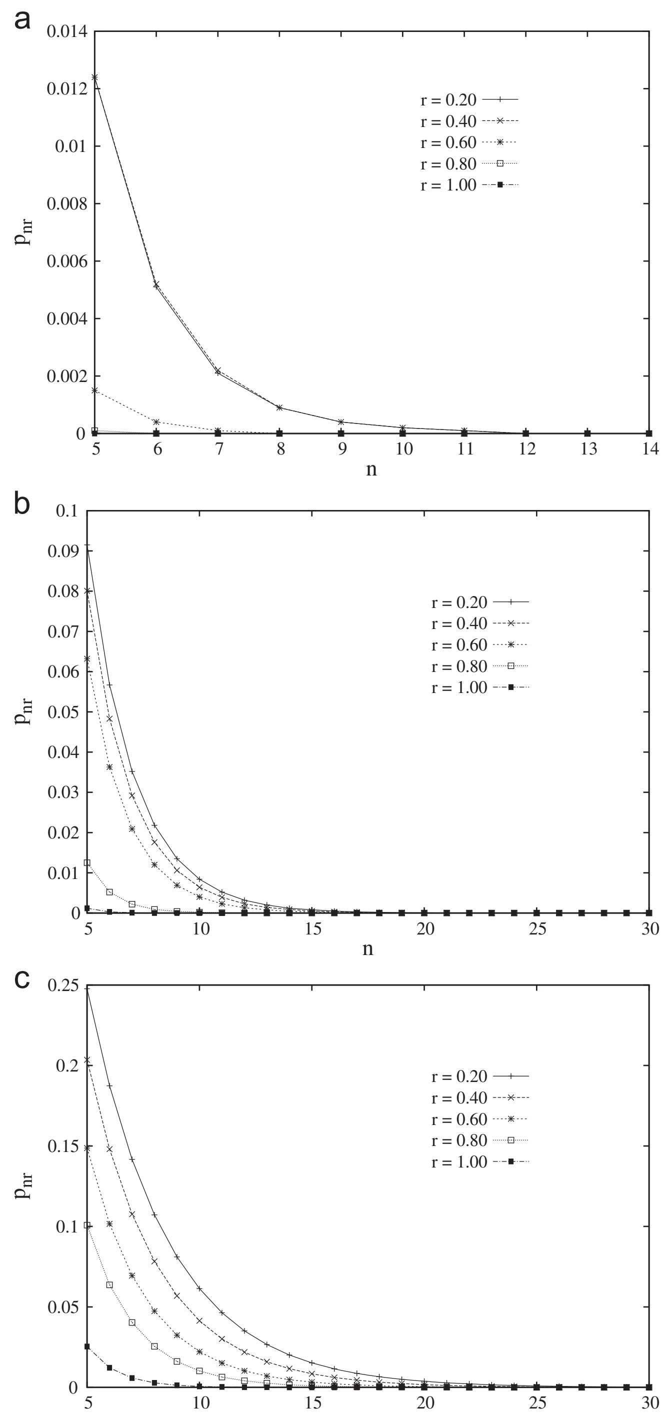

Fig. 12. Probability that a node does not receive a copy of the broadcast message as a function of a and r. (a) $a=0.40$, (b) $a=0.60$, (c) $a=0.80$.

The probability that a node does not receive any copy of the broadcast message *MSG* is

$$p^{Q}_{nr}(a,r)=(1-p^{Q}_{tx}(a,r))^{n(r)} \quad (26)$$

where $p^{Q}_{tx}(a,r)$ is the transmission probability of a node.

The effect of the value of r on the reliability of broadcasting is shown in Fig. 12 for the different values of a. The value $r=a$ represents a critical point as the probability p_{nr} drops drastically when $r>a$. For low values of a, the effect is negligible even for $r<a$. The effect is particularly pronounced for $a=0.8$ implying that an aggressive broadcasting strategy with a high value of a can only be feasible in near-perfect channel conditions. From the perspective of algorithm design, this implies that the forwarding behaviour of a node depends not only on a, its distance to the source of the overheard broadcast, but also to the farthest distance over which it can communicate reliably. We discuss more on broadcast algorithm design in dynamic channel conditions in the next section.

5.3. Insights for algorithm design

The results obtained above indicate that the broadcasting behaviour of nodes needs to adapt to the channel conditions. To accomplish such a design, it is necessary for a node to be aware of the channel conditions in its neighbourhood. However, it is not possible for a node to gain a precise estimate of its neighbourhood using the *network information* in the *BRP*. Approaches based on building connected dominating sets (CDS) typically make use of complete local or global topology information. However, in addition to the transmission overheads incurred in the dissemination of neighbourhood information, such a design is not robust to time-varying channel conditions. This necessitates the need to estimate immediate neighbourhood information. An ideal way to estimate network conditions is to make use of *network information* available in the *broadcast optimization phase* (*BOP*). As transmissions in the *BOP* are contextual to the current broadcasting process, instantaneous channel conditions can be gauged more accurately. With this in mind, we obtain insights on how to optimize node transmission behaviour based on information available in the *BOP*.

We denote by x_{BRP} and x_{BOP} the distance over which a node receives transmissions in the *BRP* and *BOP* respectively. In stateless broadcasting algorithms, the effectiveness of a node's transmission is determined primarily as a function of its EAC. As a result, if the transmission from a node i is received by a node in its *BRP* with a small x_{BRP}, then the optimal choice for the latter would be to refrain from retransmission. However, if the channel conditions are bad (i.e. r is low), a low value of x_{BRP} may not imply low EAC as there are likely to be other nodes within i's range that do not receive its transmission. Thus, it is preferable for nodes not to refrain from forwarding purely on the basis of x_{BRP}. Nevertheless, the effectiveness is still proportional to this value since a transmission received with high x_{BRP} always indicates high EAC.

In the *BOP* itself, transmissions received with high x_{BOP} do not significantly reduce the EAC of the node. However, a high value of x_{BOP} also indicates good channel conditions and vice versa. This implies that for nodes with low x_{BRP}, the discarding probability p_{disc} should be directly proportional to x_{BOP}.

6. Conclusion

In this paper, we provide a comprehensive performance analysis of stateless broadcasting algorithms. The proposed analytical model is based on the parameters derived from the information available at nodes during the different stages of broadcasting. As a result of this, the model is generic to all stateless broadcasting schemes. We identify how the *network information* at the nodes map to the broadcasting performance benefits. We use our model to obtain feasibility conditions for the algorithm to operate, given the network density and costs incurred. Finally, we show how the dynamic nature of the wireless channel can be incorporated into the existing model. We draw insights on how algorithm design can be made robust so as to adapt to varying channel conditions.

References

Banerjee A, Tantra JW, Foh CH, Yeo CK, Lee BS. Controlling route discovery for efficient routing in resource-constrained sensor networks. In: Proceedings of the IEEE consumer communications and networking conference (CCNC), short paper; January 2010, p. 233–4.

Bianchi G. Performance analysis of the IEEE 802.11 distributed coordination function. IEEE Journal on Selected Areas in Communications 2000;18(3):535–47.

Bruck J, Gao J, Jiang AA. Localization and routing in sensor networks by local angle information. ACM Transactions on Sensor Networks 2009;5(1):1–31.

Chen C, Chin-Kai H, Hsien-Kang W. A distance-aware counter-based broadcast scheme for wireless ad hoc networks. In: IEEE military communications conference (MILCOM), October; 2005. p. 1052–8.

Clark BN, Colbourn CJ, Johnson DS. Unit disk graphs. Discrete Mathematics 1990;86(December):165–77.

Foh CH, Tantra JW, Jianfei C, Lau CT, Fu CP. Modeling hop length distributions for reactive routing protocols in one dimensional MANETs. In: Proceedings of the IEEE international conference on communications (ICC), June; 2007. p. 3882–6.

Haas ZJ, Halpern JY, Li L. Gossip-based ad hoc routing. IEEE/ACM Transactions on Networking 2006;14(3):479–91.

Heissenbuttel M. Routing and broadcasting in ad-hoc networks. PhD thesis. CH-3012 Bern, Switzerland: University of Bern; June 2005.

Heissenbuttel M, Braun T, Walchli M, Bernoulli T. Optimized stateless broadcasting in wireless multi-hop networks. In: Proceedings of the 25th IEEE international conference on computer communications (INFOCOM); 2006. p. 1–12.

Huang SC-H, Wan P-J, Jia X, Du H, Shang W. Minimum-latency broadcast scheduling in wireless ad hoc networks. In: Proceedings of the 26th IEEE international conference on computer communications (INFOCOM), May; 2007. p. 733–9.

Jun T, Roy N, Julien C. Modeling delivery delay for flooding in mobile ad hoc networks. In: Proceedings of the IEEE international conference on communications (ICC), May; 2010. p. 1–6.

Kuhn F, Wattenhofer R, Zollinger A. Ad-hoc networks beyond unit disk graphs. In: Proceedings of the 1st joint workshop on foundations of mobile computing (DIALM-POMC), September; 2003. p. 69–78.

Kuo J-C, Liao W. Hop count distance in flooding-based mobile ad hoc networks with high node density. IEEE Transactions on Vehicular Technology 2007;56(3):1357–65.

Manfredi V, Crovella M, Kurose J. Understanding stateful vs stateless communication strategies for ad hoc networks. In: 17th annual international conference on mobile computing and networking (MobiCom), September; 2011.

Ni S-Y, Tseng Y-C, Chen Y-S, Sheu J-P. The broadcast storm problem in a mobile ad hoc network. In: Proceedings of the 5th annual international conference on mobile computing and networking (MobiCom); 1999. p. 151–62.

Oikonomou K, Kogias D, Stavrakakis I. Probabilistic flooding for efficient information dissemination in random graph topologies. Computer Networks 2010;54(10):1615–29.

Perkins C, Belding-Royer E, Das S. Ad hoc on-demand distance vector (AODV) routing; July 2003. Available: 〈http://www.ietf.org/rfc/rfc3561.txt〉.

Shah-Mansouri H, Bahramian S, Pakravan MR. Performance analysis of flooding over CSMA in wireless ad-hoc networks. In: Proceedings of the IEEE 19th international symposium on personal, indoor and mobile radio communications (PIMRC), September; 2008. p. 1–5.

The UCB/LBNL/VINT network simulator-ns (version 2). VINT project [online]. Available: 〈http://www.isi.edu/nsnam/ns〉.

Viennot L, Jacquet P, Clausen T. Analyzing control traffic overhead versus mobility and data traffic activity in mobile ad-hoc network protocols. Wireless Networks 2004;10(4):447–55.

Viswanath K, Obraczka K. Modeling the performance of flooding in wireless multihop ad hoc networks. Computer Communications 2006;29(May):949–56.

Wang D, Abouzeid A. Link state routing overhead in mobile ad hoc networks: a rate-distortion formulation. In: Proceedings of the 27th IEEE international conference on computer communications (INFOCOM), April; 2008. p. 1337–45.

Wieselthier JE, Nguyen GD, Ephremides A. Energy-efficient broadcast and multicast trees in wireless networks. Mobile Networks and Applications 2002;7(6):481–92.

Williams B, Camp T. Comparison of broadcasting techniques for mobile ad hoc networks. In: Proceedings of the 3rd ACM symposium on mobile ad hoc networking and computing (MOBIHOC), June; 2002. p. 194–205.

Williams B, Mehta D, Camp T, Navidi W. Predictive models to rebroadcast in mobile ad hoc networks. IEEE Transactions on Mobile Computing 2004;3(3):295–303.

Wu W, Du H, Jia X, Li Y, Huang C-H. Minimum connected dominating sets and maximal independent sets in unit disk graphs. Theoretical Computer Science 2006;352(1):1–7.

Zhang H, Jiang ZP. Performance analysis of broadcasting schemes in mobile ad hoc networks. IEEE Communications Letters 2004;8(12):718–20.

Zhou N, Wu H, Abouzeid AA. Reactive routing overhead in networks with unreliable nodes. In: Proceedings of the 9th annual international conference on mobile computing and networking (MobiCom), September; 2003. p. 147–60.